Equivalent Expressions: Expanded and Factored Forms

Distributing to expand, factoring out the greatest common factor, combining like terms, and testing equivalence by substitution

One idea, read in two directions. The distributive property says $a(b+c) = ab + ac$. Read left to right it **expands**; read right to left it **factors**. Two expressions are **equivalent** when they give the same value for *every* number you substitute — so a single substitution can prove two expressions are *not* equivalent, but it never proves that they are.

Directions.

- Decide first which direction the problem asks for: expand, factor, or collect like terms. The problems do not stay in one mode.
- Factor *completely* — if the terms left inside the parentheses still share a factor, you are not finished.
- Write your final expression on the answer line.

1. Expand: $3(x+4)$

Write the multiplication you do for each term inside the parentheses before you simplify.

Answer: ________________

2. Factor: $5x + 15$

Name the greatest common factor of the two terms first, then write the expression as that factor times a sum.

Answer: ________________

3. Expand: $-2(x-7)$

The multiplier is negative and the expression inside is a difference. Show both products with their signs.

Answer: ___________________

4. Write an equivalent expression with as few terms as possible:

$$4x + 3x + 6$$

Say which terms are like terms and why the remaining term cannot join them.

Answer: ___________________

5. Ana wrote $3(x+4) = 3x + 4$.

(a) Substitute $x = 5$ into $3(x+4)$ and find its value. Put that value on the answer line.

(b) Substitute $x = 5$ into Ana's expression $3x + 4$. She gets 19.

(c) The two values disagree, so the expressions are not equivalent. Describe Ana's mistake in one sentence.

Answer: ____________________

6. Factor completely: $12x + 18$

Someone factors this as $2(6x+9)$ and stops. Explain why that is not finished, then give the fully factored form.

Answer: ____________________

7. Expand: $-4(2x - 3y + 1)$

There are three terms inside the parentheses, so there are three products. Keep track of every sign.

Answer: ________________

8. Write an equivalent expression with as few terms as possible:

$$2(x + 3) + 4x$$

You cannot combine anything until the parentheses are gone. Show the expanding step and the collecting step separately.

Answer: ________________

9. Dev claims that $4(2x-5)$ and $8x-5$ are equivalent.

(a) Substitute $x=4$ into $4(2x-5)$ and put that value on the answer line.

(b) Dev's expression $8x-5$ gives 27 at $x=4$.

(c) Are the two expressions equivalent? Say how the single substitution settles it, and write the correct expansion of $4(2x-5)$.

Answer: __________________

10. Factor completely: $8x-20y$

Two different letters appear, so no variable is common to both terms. Find the greatest common *number* and write the factored form.

Answer: __________________

11. Expand and simplify:

$$-3(4 - 2x) + 2(x + 6)$$

Expand each pair of parentheses first, then collect like terms. Something surprising happens to the constant terms — say what and why.

Answer: ________________

12. **Challenge.** A club is printing $18x + 24$ stickers.

(a) Factor $18x + 24$ completely, so the club can see how many equal bundles it can make. Put the factored expression on the answer line.

(b) Substitute $x = 2$ into your factored form and give its value.

(c) Substitute $x = 2$ into the original $18x + 24$. Does it agree with part (b)? Explain in one sentence why agreeing at $x = 2$ is good evidence but not a proof of equivalence.

Answer: ________________